जांच समिति की रिपोर्ट

आग की घटना आई ओ सी एल, हल्दिया रिफाइनरी

घटना की तिथि 21st दिसम्बर, 2021

INVESTIGATION COMMITTEE REPORT

ON FIRE INCIDENT AT IOCL - HALDIA REFINERY

DATE OF INCIDENT: 21st DECEMBER, 2021



तेल उद्योग सुरक्षा निदेशालय

OIL INDUSTRY SAFETY DIRECTORATE

पेट्रोलियम एवं प्राकृतिक गैस मंत्रालय, भारत सरकार

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OIL INDUSTRY SAFETY DIRECTORATE

EDS: OISD: ACCI/ 06 **Dt: 14.02.2022**

Sub: INVESTIGATION COMMITTEE REPORT ON FIRE INCIDENT AT IOCL–HALDIA REFINERY ON 21.12.2021.

1.0 INTRODUCTION

MSQ units were under planned shutdown since 13.12.2021 and various shutdown jobs were being carried out after normal shutdown and hand over of the units to Maintenance.

Indian Oil Corporation Limited, Haldia Refinery (IOCL-HR) vide FIR dated 21.12.2021 (Ref. **Annexure-1**), informed that at around 14:50 hrs on 21st December 2021, a fire had occurred in the area of MSQ Block (Isomerization) of the Refinery while grating/ structure cutting job was being carried out in the first platform of Stabilizer area.

The fire was put-off by applying water by personnel working in the vicinity and subsequently by Firefighting crew. The duration of the fire was less than 15 minutes as per FIR copy received from IOCL-HR.

There were 3 fatalities and 44 injuries of contract personnel due to the incident as reported in the FIR. However, IOCL subsequently revised the casualty figure to 3 fatalities and 41 injuries. At the time of release of this report, total fatalities increased to 5. There was no injury to any of the IOCL personnel.

1.1 To investigate the incident, a committee comprising of the below mentioned officers was constituted by ED, OISD, vide letter ref. OISD/EDS/ACCI-05 dated 22/12/2021 (**Annexure-2**).

1. Shri Udit Nandi, CGM (Safety), HPCL- Visakh Refinery
2. Shri S Giridharan, GM (Maint.), BPCL- Kochi Refinery
3. Shri Jamunalal Rout, Dy. Director, PNGRB
4. Shri Harendra Kumar Yadav, Addl. Director (P&E), OISD

The terms of reference of the committee is as under:

- Investigate the incident
- Identify the lapses and root cause(s) of the incident;
- Recommend corrective action to avoid recurrence of such incidents.

2.0 INVESTIGATION MODALITIES

The committee members visited the incident site from 23rd to 24th December, 2021 and carried out the investigation following the below noted methodology:

- Visit to MSQ Plant.
- Visit to Field Room
- Visit to Main Control Room of MSQ Block.
- Visit to Fire Station
- Interaction with concerned stakeholders as mentioned below:

S. No.	Name of the Person	Designation	Company
1	Shri A Sanyal	CGM(TS&HSE)	IOCL
2	Shri D Dutta	CGM (T)	IOCL
3	Shri I. Daniel Raj	DGM (HSE)	IOCL
4	Shri Shubham Jain	AMPS (Technical Service)	IOCL
5	Shri Mridul Deka	ITM	IOCL
6	Shri Ashok Kumar Raghuvanshi	CMFS	IOCL
7	Shri Panner Selvam Ponnusamy	SPNE	IOCL
8	Shri Golap Thakuria	SPNE	IOCL
9	Shri Amit Binwal	SPNE	IOCL
10	Shri Shahrukh Husain Ansari	AMPN	IOCL
11	Shri K.S. Paul	DGM (PN)	IOCL
12	Shri S.M. Raut	SMFS	IOCL
13	Shri Jaydip Chattaraj	SMNM	IOCL
14	Shri Sreekuttan M.S	AMFS	IOCL
15	Shri Neelabh Gupta	SF & SO	IOCL
16	Shri Saurabh Mukherjee	SPNM	IOCL
17	Shri Samaresh Biswas	MNMEL	IOCL
18	Shri Mulakala Venkata Niranjan	AM (Mechanical Manager)	IOCL
19	Shri Himagna Sen	AM-ML (Mechanical Maintenance)	IOCL

S. No.	Name of the Person	Designation	Company
20	Shri Naveen Kumar	SIPM	IOCL
21	Shri Mukesh	FO (Field Operator)	IOCL
22	Shri Ankit Chaturvedi	FO (Field Operator)	IOCL
23	Shri Susanta Das	FO (Field Operator)	IOCL
24	Shri Kanak Kanti Sahoo	FO (Field Operator)	IOCL
25	Shri Krishnanand Kharwal	FO (Field Operator)	IOCL
26	Shri Saroj Dey	FO (Field Operator)	IOCL
27	Shri Rangan Datta	FO (Field Operator)	IOCL
28	Shri Ranjit Pratap Singh	FO (Field Operator)	IOCL
28	Shri Devendra Dhar Dwivedi	FO (Field Operator)	IOCL
29	Shri MD. Mahasin Khan	FO (Field Operator)	IOCL
30	Shri Rajib Kuiti	FO (Field Operator)	IOCL
31	Shri Sanjoy Mandal	FO (Field Operator)	IOCL
32	Shri Daruknath Mandal	FO (Field Operator)	IOCL
33	Shri Dipanjan Mal	FO (Field Operator)	IOCL
34	Shri Sanjay Sahai	Site Incharge	M/s Shilpi
35	Shri Konshik Gole	Project Manager	M/s Shilpi
36	Shri Zanaklal	Safety Officer	M/s Shilpi
37	Shri Rishin Raj	Supervisor	M/s Shilpi

Note: *Injured personnel could not be contacted for interaction as all were under treatment at Kolkata.*

- Review of documents viz. DCS logbooks/ Field logbooks/ maintenance register/ Work permits/ registers in the control room/ Operating manuals/ F&S logbook/ Contract document/ SOPs.
- Review of CCTV footage from camera DHDS-508, DHDS-509, DHDS-510, DHDS-701, DHDS-515 etc.
- Review of IOCL's internal investigation report findings.

3.0 BACKGROUND

Haldia Refinery is situated on the banks of river Hoogly in the Haldia town of East Midnapore District in West Bengal and is about 130 km from Kolkata by road. It is a combination of both fuel and lube refinery. Apart from normal fuel distillates like, LPG (Liquefied Petroleum Gas), MS (Motor Spirit), ATF (Aviation Turbine Fuel), Kerosene, Diesel etc., the refinery also produces various grades of Lube Oil Base Stocks (LOBS) for production of different grades of lubricating oils.

Refinery was commissioned in the year 1975 with a crude processing capacity of 2.5 MMTPA. The LOBS production facility was commissioned in 1977. With time, the refinery has been revamped in phases to the present capacity of 8.0 MMTPA. The Refinery processes approx. 70% high Sulphur and 30% low Sulphur crude.

The MSQ block comprising of the NHT (unit no. 85), ISOM (Unit 86), Prime G (unit no. 87) was commissioned in year 2005 and Prime-G was revamped in 2019 for BS-VI projects. Last turnaround of the MSQ Units was taken in November 2017.

The manning of MSQ units on the day of the incident was as given below: - .

	Normal Manning	Manning on 21.12.2021
Field operator (JEA)	4	7
Panel	1	1
Shift In-charge (SIC)	1	2 *
Contract personnel		Approx. 340

** Two Production officer deployed in shift; Single SIC not designated .*

4.0 INCIDENT DESCRIPTION AND CHRONOLOGY OF EVENTS

MSQ block comprises of VDU-II, Prime-G, Isomerization and NHT units. These units were having the planned shutdown as below:

Unit	Shut down duration
NHT	24 days (13th Dec – 5th Jan 2022)
ISOM	24 days (13th Dec – 5th Jan 2022)
Prime-G	52 days (1st Dec – 21th Jan 2022)
VDU-2	4 days (13th Dec – 16th Dec 2021)

Isomerization unit shut down activities were going on since 13.12.2021 as per the shutdown work plan along with other units of the MSQ block. As per IOCL-HR:-

a. The ISOM unit was (including the Stabilizer column, 86-C-01 circuits) made hydrocarbon free by N2 purging, water flushing and steam purging.

b. Stabilizer column (86-C-01) was blinded on 20.12.2021. On 21.12.2021, stabilizer reboiler (86-E-06) inlet flanges were being opened to facilitate blinding for hydrotest of reboiler inlet circuit as one of the shutdown jobs. (Refer Fig. 1).

c. During the flange opening activity of Reboiler inlet line, Naphtha along with water started coming out through the opening.

d. At 14:50 hrs, there was an explosion followed by fire on the first floor of Unit 86 (ISOM) at the Stabilzer column area; Fire continued for less than 15 minutes as per FIR intimation by IOCL-HR.

   Fire call was received from MCP 609 and simultaneously message was received on a walkie-talkie regarding fire in MSQ unit. Subsequently fire call received from MCP 640, 610, 625, 611, 620, 612, 606, 751, 736, 622 and 644.

e. Initial fire fighting activities started with water spray from nearby ground monitors and fire hydrants by contractor personnel. Fire tender-1 from the main fire station and Fire tender-4 from the satellite fire station reached the site and responded in fire fighting activities.

f. Simultaneously rescue operation was started and casualties were shifted to Occupational Health Centre for first aid by ambulance and other available vehicles. Subsequently injured were shifted to Hospitals at Kolkata.

g. Shutdown manpower from the surrounding areas were evacuated and affected area was cordoned off.

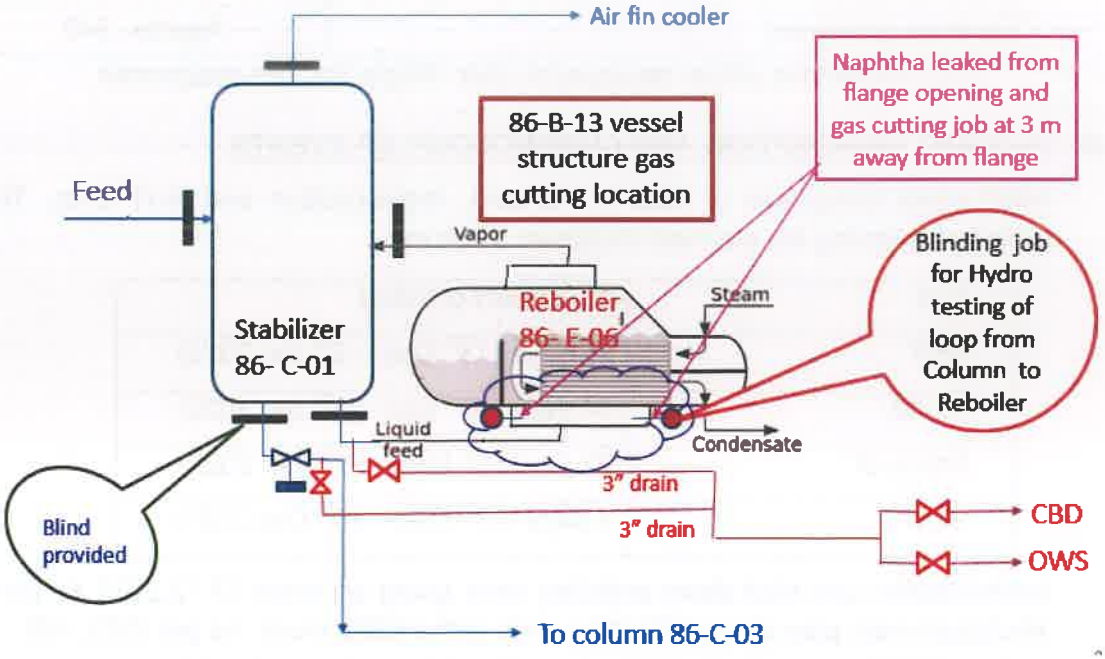
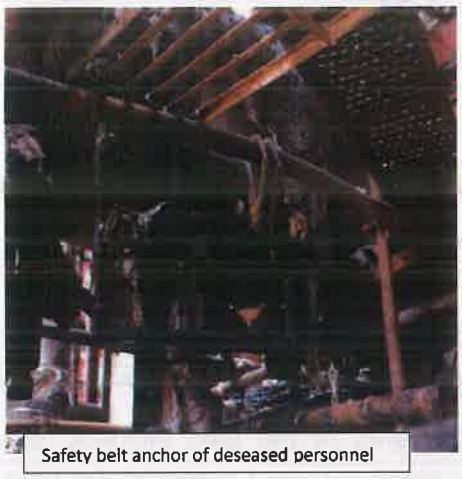


Fig.1 (Site activities being carried out on the day at the time of incident)

5.0 Scene of Accident as noted during investigation team's visit to site:

1) At the time of site visit, it was observed that no activities were going on as PESO has given order not to carry out any job in the unit.
2) CISF had cordoned off the unit and all personnel entry to the unit were being recorded in a register.
3) Stabilizer column (86-C-01) shell flanges including reboiler inlet and outlet lines were noted in blinded condition.
4) Considerable material damage at incident site was noticed as can be seen from the photographs below. It was observed that the anchoring hooks of safety belt of the deceased along with tools and tackles (wrench, spanner, hammer etc.) were found still present at site. Tools and tackles were found to be of normal metallic design (i.e. not of 'Non sparking' type).



Safety belt anchor of deseased personnel





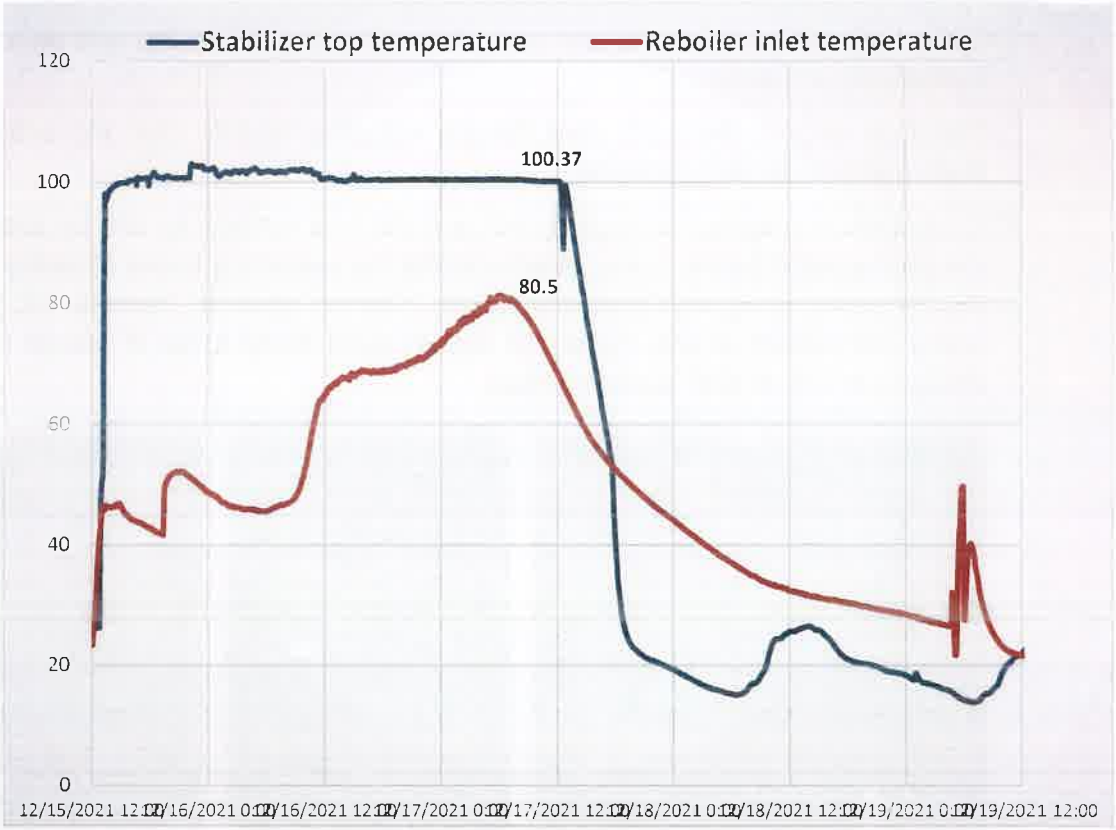
5) Details of deceased and injured personnel is given in the Annexure-3.

6.0 Observations by investigation committee:

6.1 Observations based on DCS trends (Ref. Annexure-4) and CCTV footages

1) Isomerization unit feed was reduced at 2200 hrs. on 12.12.2021 and feed cut out was done at 00:00 hrs on 13.12.2021.
2) Steaming of Stabilizer column (86-C-01) was carried out from 15.12.2021 (12:45 hrs.) to 17.12.2021 (12:55 hrs.) as per DCS history data (Ref. stabilizer top temperature 86TI0614-PV).
3) Column 86-C-01 bottom level remained low between 13.12.2021 to 19.12.2021 in the range of 4 – 5%. On 19.12.2021 level increased to 104% at 13:50 hrs. and again came down to 0% at 16:35 hrs.

4) Reboiler 86-E-06 inlet temperature reached maximum of 81°C (TI-0616 tag) on 17.12.2021 during the steam purging phase. It is inferred that reboiler circuit did not get steam purged adequately.



5) It was noted that for the sequence of events of the said incident, there was difference w.r.t. timing (varying between days/ minutes) amongst the time record of DCS clock, CCTV footages, alarm server time.

6) On 24.12.2021, in presence of investigating team and IOCL internal investigation team, water was poured in the reboiler vapour return line by opening a thermo-well flange (86TI0604). Naphtha was observed coming out from the reboiler inlet flanges at reboiler bottom (which were in open condition).

   Sample was collected and sent to laboratory for testing; Same was found to be having Light Naphtha properties (IBP 52°C / FBP 75°C).

   During these activities, reboiler and stabilizer column common drain line was found choked. This observation indicated that naphtha was still remaining inside the reboiler vessel/ circuit, although steaming / N2 purging activities had been done as per shutdown procedure.



Flange joint from Stabilizer column to reboiler with blank, Referred in 4.0b)



Reboiler inlet flange: Gap created by opening the bolts, Referred flange in para 6.1.6)

7) From alarm summary, it was observed that HC gas detector 82006AI0301A got activated for high-high alarm at 13:10:27 hrs. on 21.12.2021. IOCL should ensure that all the critical alarms are given due attention and necessary actions/ checking of root cause for such alarms are addressed accordingly.

8) On CCTV camera (tag no. DHDS-515), record was not available after about 14:00 hrs. As stated by Instrumentation department, subject CCTV outage was due to non-availability of power; This was in contradiction with the statement of the Electrical Maintenance department that additional power source was provided before UPS outage and UPS was also handed over to Instrumentation at 05:30 hrs on 21.12.2021). IOCL-HR may review the fact.

9) No historical trend was available for the HC detectors (86-AE-1003, 86-AI-0701 near pump 86-PM-06B) in DCS system. All the gas detector should have historical trend.

6.2 Observations based on the interaction with personnel:

1) The committee was informed during site visit that 14:00 hrs was the operator shift change-over time. Seven field operators were deployed in each shift in the MSQ plant during M&I. On the incident day of 21st December 2021, 3 nos. of field operators were continued in the evening shift and rest four operators were to be relieved.

2) On that day (i.e. 21/12/2021), a mock drill was conducted from around 11:00 to 14:15 hrs in another refinery process unit SRU; Accordingly, vehicles entry to the refinery was restricted resulting in delay in arrival of four Evening Shift operators for change-over. However, during interactions, it was revealed that 4 field operators of the morning shift left the plant prior to arrival of their relievers

and one field operator (who stayed back) went for taking lunch at the time of the incident. Two other operators available in the MSQ block were working in NHT section (Unit no 85). So, no field operators were available in ISOM Unit (Unit no 86) to supervise the maintenance activities; Such crucial administrative gaps may be reviewed by IOCL-HR management.

3) IOCL personnel were not present in unit no. 86 for supervision of the maintenance jobs being carried out by contractor personnel. Regular supervision could have stopped all maintenance activities (including the hot job) and evacuate the manpower deployed, immediately on observing Naphtha leakage.

4) Documentary evidence regarding head count of the people of MSQ block after the accident could not be furnished by the refinery.

5) There were few fire incidents/ near misses (noted below) prior to the accident emphasising presence of hydrocarbon in the shutdown plants and inadequate actions thereof.

   a) Fire incident occurred in the refinery on 15.12.2021, during cutting of 2-inch LPG reprocessing line in FCC unit (under shutdown). As per refinery F&S department record, the cause of fire was due to trapped hydrocarbon in the lines as a result of improper flushing.

   b) Fire call (MCP 609, 610) was received on 19.12.2021 at 09:13 hrs about a fire in a culvert between MSQ and FCCU. Again at 17:13 hrs, Fire call received on the same day and from the same location.

   c) On 20.12.2021, the Fire & Safety officer noticed naphtha dripping from the blinded flange of process line from 86-C-03 column. However, shift officer, during interaction in presence of fire and safety, confirmed that naphtha leakage was from the stabilizer (86-C-01) column feed inlet shell flange and not from 86-C-03. This leakage was due to presence of naphtha in the circuit and inadequate tightening of flanges. Leak was stopped by bolt tightening.
   This instance of naphtha leak was highlighted by F&S officer to all the concerned by e-mail and was discussed in the shutdown meeting of 20.12.2021 (Annexure-5 – F&S statement). This indicates that refinery was aware of presence of naphtha in the isomerization unit circuits.

   The investigating team is of the opinion that such `Unsafe Acts' and `Unsafe Conditions' should be analysed critically and preventive steps be taken by the Refinery team to avoid occurrence of major incidents.

6) No record was produced w.r.t. the list of daily clearances given by Production against the valid work permits. The Mechanical maintenance section stated that daily clearance was obtained against the permit No 904021081364/

9040CL21075294 for flange opening of the stabilizer reboiler but the Production engineers stated in contradiction. The fact could not be established as the file containing the clearance was found burnt in the incident. It was informed that during M&I, as the volume of jobs were higher than usual, daily clearance in work permits were given manually without maintaining any record of list of clearances issued.

7) All the witnesses have informed and CCTV footage of DHDS-508 camera has confirmed that there was an explosion followed by fire. The intensity of explosion was felt even at 100 m away.

8) It was noted that operation of the incident control room was not activated subsequent to the subject emergency. IOCL stated that emergency control room was not activated as fire was less than 15 minutes. However, the incident involved an explosion followed by fire and as per IOCL-HR ERDMP chapter 10.1e. (Major Disaster Scenarios), fire & explosion in unit area is categorized as Level-2. The organization's emergency protocol should be put into action from the initial stages as per chapter 16 of ERDMP.

6.3 Additional / other observations and Analysis related to the Incident

The committee noted the following additional relevant information based on site visit (23rd & 24th December-2021), discussion and analysis of various records/ documents: -

1) The chlorine guard bed (86-B-13) grating, adjacent to the incident area, was being cut for removal of the vessel from the Isomerization Unit; This activity was already in progress when the parallel job of blinding of Stabilizer Reboiler bottom was started on 21.12.2021;

2) Refinery could not produce approval of the statutory authority (PESO) for removal of the chlorine guard, which is essential for any change of Layout as per Petroleum Rules 2002.

3) It was also observed that the MSQ block units (namely Prime-G unit, Isomerization Unit, NHT unit, VDU-II) were not meeting the inter distance requirement as per OISD-STD-118 Table-1; So these units are to be considered as single block and their shutdown to be taken simultaneously as per clause 5.1 c) of OISD-STD-118. However, this philosophy was not followed in the current shutdown. The Prime-G unit was shut down for M&I on 1st December 2021 whereas the Isomerization unit and the adjacent VDU-II were operating till 12th December 2021. This is also a violation of commissioning permission issued by statutory authority (PESO) and OISD-STD-118 Table-1.

4) ISOM unit was observed to be congested with scaffolding without having proper escape routes for the large numbers of contract workmen deployed in the unit for maintenance activities. The Operation, Maintenance and Fire & Safety team should have carried out the Hazard Identification and Risk Assessment (HIRA) for such high number of manpower deployed in the unit. The job scheduling should commensurate with the available space for working and escape provisions.





5) During interaction, it was revealed that due to level-III mock drill in the morning shift (as noted in para 6.2.2), the jobs were stopped for about 3 hrs; So more people were there in the second shift to finish the scheduled jobs for the day. The casualty/ injury might have been less if proper job scheduling was done, thereby staggering the manpower deployment.

6) It was observed on 23.12.2021 by investigation team that in VDU-2, hot job (Hot permit no. 9040H2113262) was going on for carrying out welding job at height without proper booth (enclosure); Also, some personnel were found not wearing the boiler suit / apron.

7) Column 86-C-01 & 86-C-03 overhead AFCs (86EA01, 86EA 02 A/B) have not been provided with water sprinkler system. As per OISD-STD-116 clause no. 4.2.5, AFCs in hydrocarbon service located above pipe racks / elevated location shall be provided with water sprinkler system.

8) During visit to the MSQ operator room, the Log book was checked and it was noticed that neither the Shift-in-charge nor the Operators had signed in the log sheet. IOCL senior officers should be involved in cross checking.

9) From the list of injured personnel, it is observed that 50% of the workers were in the twenties (lowest age being 19 yrs.). Also, two safety supervisors for 340

contract personnel appeared to be low. Experience criteria of these workers and number of safety supervisors need a thorough review.

10) As per the Work Order no. 27152457 for shutdown activities of MSQ units; skilled manpower shall be verified by Engineer-in-charge before deployment on the job. Minimum one Safety-in-charge and two safety supervisors to be provided in each shift by the Contractor. No job specific PPEs was mentioned in the work order.

11) The total casualty reported in FIR was 47 (fatality-3 & injured 44) , the same reported in ERDMP Schedule-VI (Incident Reporting Format) was 41 (fatality-3 & injured 38) and IOCL-HR subsequently corrected the total casualty to 44 (fatality-3 & injured 41). IOCL-HR could not produce the name of the three injured workmen who were released on the same day after preliminary treatment.

12) Two production engineers were deployed in each shift during M&I of MSQ unit. However, none of them were designated as Shift-in-charge of the unit. In absence of identified in-charge of shift, it was difficult to ensure proper emergency command and control especially during odd hours.

7.0 PROBABLE REASONS OF FAILURE / ROOT CAUSE

1) Inadequate flushing of the Reboiler circuit and lack of site supervision by IOCL during execution of non-routine activities were the primary cause of incident. Final checking for ensuring 'nil' hydrocarbon presence at reboiler bottom was not carried out properly.

2) Inadequate risk awareness/ sensitization among the contract workmen/ IOCL employees contributed to the incident. As soon as Naphtha started spilling out from the reboiler bottom flange, concerned personnel could have alerted all people in the vicinity (particularly the ones involved in grating cutting job of chlorine guard vessel).

3) Due diligence w.r.t. suitable action against early sign/ unsafe condition was not displayed after noticing the Naphtha leak from 86-C-01 feed inlet blinding shell flange on 20th December 2021 at 10:30 hrs for making the stabilizer bottom circuits hydrocarbon free.

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8.0 CONCLUSION

1) Based on the observations and discussions noted above, it is concluded that source of fuel for the fire was the spilled out residual Naphtha present in the reboiler (86-E-06) and connected piping during opening of reboiler inlet flanges while carrying out the blinding activity.
2) The source of ignition most likely was from (a) the hot job going on in the vicinity (gas cutting job of grating for removal of chlorine guard 86-B-8) or (b) spark generated due to use of normal metallic tools.

9.0 RECOMMENDATIONS

Based on the findings, the committee hereby proposes the following recommendations:

1. Adequacy of Standard Operating Procedure (SOPs) for flushing to be reviewed to ensure incorporation of equipment wise flushing plan and to include physical checking/ verification to confirm system is hydrocarbon free.
2. Hazard Identification and Risk Assessment (HIRA) shall be conducted by Operation, Maintenance and Fire & Safety team before start of non-routine activities.
3. Maintenance crew engaged for various shutdown jobs to be properly made aware (through supervisory control, tool box talks etc.) about the potential job hazards, Do's & Don'ts in case of any unforeseen abnormal condition like oil/ gas leakage in and around the job site. They should be sensitized about their individual role as a `Safety watchdog' for protection of self and others in the vicinity of work.
4. Deployment of skilled manpower suitable for the respective job and commensurate with contract condition to be ensured. It is suggested to include suitable clauses in the shut-down tender document to ensure that only qualified and experienced workman are entering into refinery premises for shutdown job.
5. In line with OISD-STD-155, it is recommended that IOCL shall carry out hazard and risk assessment and develop a PPE policy. Fire retardant/ flame resistant coverall shall be included as mandatory PPE in the policy based on risk assessment.

   Implementation of PPE policy should also be ensured for contractor's workforce, through incorporation of suitable requirements in the tender/ contract.
6. OWS and CBD drain lines of equipment are to be always maintained de-choked/ operational to facilitate proper draining during steaming & flushing and ensuring positive confirmation that all hydrocarbons are drained.

7. Sufficient numbers of Safety supervisors shall be ensured all the time during the shutdown/ start-up activities of the units. Critical non-routine activities shall be carried out under supervision by IOCL along with fire watch assistance.
8. A multidisciplinary team (MDT) should be constituted to carry out the surprise checks during shutdown activities, to identify the unsafe acts and unsafe conditions. Due diligence to be accorded for review/ analysis of unsafe acts/ conditions reported by MDT, fire & safety team and operational personnel to take immediate corrective action.
9. Availability of CCTV cameras installed at all units to be reviewed and suitably augmented (if required) to record the critical site activities. Use of Artificial Intelligence (AI) equipped CCTV system may be explored, so that there is auto-orientation of the cameras towards the site of abnormality/ Fire.
10. There may be some instance where CCTV cameras and HC Gas monitoring system (GMS) may not be available to cover critical maintenance activities. To take care of such instance, IOCL to explore the feasibility of providing portable CCTV camera and GMS sensor.
11. Date/ time difference between alarm server, DCS system, CCTVs to be rectified and all time to be synchronized to a master clock (GPS time synchronization) to facilitate accurate interpretation of the recorded data.
12. Man-to-man relieving in the unit to be ensured for proper handover and takeover of duties.
13. Logging, entry of daily activities and sequence of operation performed in respective areas for all critical activities are to be ensured in the log book. Logbooks shall be reviewed and countersigned by the role holder officers on daily basis.
14. Scaffolding of erection for shut down unit shall be reviewed for easy access and escape route. Number of simultaneous activities to be regulated based on the site condition/ congestion.
15. Prior approval of the statutory authority (PESO) shall be taken for any change of approved layout as per Petroleum Rules 2002.
16. MSQ block units (including VDU-II) shall be operated on simultaneous shutdown philosophy as a single block, considering that the inter distance criteria of OISD-STD-118 is not complied. Single block philosophy, wherever established, to be strictly adhered in other process units during shutdown.
17. All the detectors (hydrocarbon, toxic etc.) shall be configured in the history trend of DCS system.



18. Incident control center operation to be ensured appropriately as per ERDMP.
19. All the Air Fin Coolers (AFCs) in hydrocarbon service located above pipe racks/ elevated location shall be provided with water sprinkler system in line with OISD-STD-116 clause no. 4.2.5.
20. All the observations/ recommendations submitted through the reports of IOCL Internal Investigation, PESO, other investigating authorities are to be reviewed and suitably addressed.

Udit Nandi	S Giridharan	Jamunalal Rout	Harendra Kumar Yadav
CGM (Safety) HPCL, Visakh Refinery	GM (Maint.) BPCL, Kochi Refinery	Dy. Director (Tech) PNGRB	Addl. Director (P&E) OISD, NOIDA

ABBREVIATIONS:

AFC : Air Fin Cooler	NDT : Non-destructive testing
CCTV: Closed Circuit Television Camera	NHT: Naphtha Hydro-treater
CR : Circulating Reflux	OISD : Oil Industry Safety Directorate
CS : Carbon steel	OWS : Oily Water Sewer
DCS : Distributed Control Systems	Prime G: Gasoline Treatment
FT : Fire Tender	PPE: Personal Protective Equipment
IWL : Inspection Work List	PMI : Positive material identification
HIRA: Hazard Identification and Risk Assessment	TA : Turn Around
HOD : Head of Department	TPI : Third Party Inspection
MSQ : Motor Spirit Quality	UT : Ultrasonic thickness
MOV : Motor Operated Valve	VDU : Vacuum Distillation Unit

ANNEXURES

Annexure -1: FIR by IOCL- HR

Annexure -2: OISD Committee Constitution note

Annexure -3: List of injured personnel

Annexure -4: DCS data

Annexure -5: Statement of F&S.